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Adaptive Noise-Cancelling Earplug with Ambient Passthrough

ABSTRACT

An earplug device includes an outward-facing microphone, an inward-facing microphone positioned near the ear canal, a digital signal processor, and a speaker driver. The processor generates an anti-noise waveform from the outward microphone signal to attenuate ambient sound, while selectively passing through specific frequency bands, such as human speech, based on a user-selected operating mode.

BACKGROUND

Passive earplugs attenuate all ambient sound uniformly, preventing a wearer from hearing important sounds such as nearby speech or alarms. Existing active noise-cancelling devices are typically built into over-ear or in-ear headphones rather than compact earplug form factors suitable for sleep or prolonged wear.

SUMMARY

In one embodiment, the earplug housing contains an outward facing microphone that captures ambient sound, a digital signal processor that computes an inverted waveform for active noise cancellation, and a speaker driver that emits the combined cancellation and passthrough signal into the ear canal. A pressed button on the housing cycles between a full-attenuation sleep mode and a speech-passthrough conversation mode.

CLAIMS

1. An earplug apparatus comprising: an outward-facing microphone; a digital signal processor coupled to the microphone and configured to generate an anti-noise signal; and a speaker driver configured to emit the anti-noise signal into an ear canal. 2. The apparatus of claim 1, wherein the digital signal processor is further configured to selectively pass through a speech frequency band in a conversation mode. 3. The apparatus of claim 1, further comprising a manual control for switching between the conversation mode and a full-attenuation mode.